SESSION 3 LAB EXPERIMENTS

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1.

```
def find_paths(m, n, N, i, j):
    dp = [[0] * n for _ in range(m)]
    dp[i][j] = 1
    result = 0
    MOD = 10**9 + 7

    for _ in range(N):
        next_dp = [[0] * n for _ in range(m)]

        for r in range(m):
            for c in range(n):
                ways = dp[r][c]
                if ways == 0:
                    continue

                for dr, dc in [(1, 0), (-1, 0), (0, 1), (0, -1)]:
                    nr, nc = r + dr, c + dc

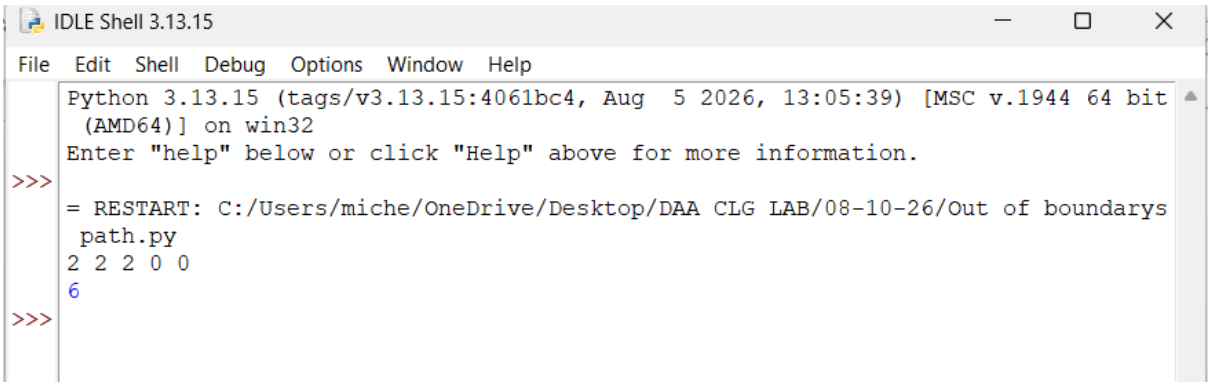
                    if nr < 0 or nr >= m or nc < 0 or nc >= n:
                        result = (result + ways) % MOD
                    else:
                        next_dp[nr][nc] = (next_dp[nr][nc] + ways) % MOD

        dp = next_dp

    return result

m, n, N, i, j = map(int, input().split())
```

print(find_paths(m, n, N, i, j))



2.

```
def rob(nums):
    def helper(arr):
        prev2 = 0
        prev1 = 0

        for money in arr:
            current = max(prev1, prev2 + money)
            prev2 = prev1
            prev1 = current

        return prev1

    if len(nums) == 1:
        return nums[0]

    return max(helper(nums[:-1]), helper(nums[1:]))

nums = list(map(int, input().split()))
print(rob(nums))
```

```
= RESTART: C:/Users/miche/OneDrive/Desktop/DAA CLG LAB/08-10-26/House Robber II.
py
2 3 2
3
>>>
```

3.

```
def climb_stairs(n):
    if n <= 2:
        return n

    a, b = 1, 2

    for i in range(3, n + 1):
        a, b = b, a + b

    return b

n = int(input())
print(climb_stairs(n))
```

```
= RESTART: C:/Users/miche/OneDrive/Desktop/DAA CLG LAB/08-10-26/Climbing Stairs.
py
4
5
>>>
```

4.

```
def unique_paths(m, n):
    dp = [1] * n

    for i in range(1, m):
        for j in range(1, n):
            dp[j] += dp[j - 1]
```

```
    return dp[-1]


m, n = map(int, input().split())
print(unique_paths(m, n))
```

```
= RESTART: C:/Users/miche/OneDrive/Desktop/DAA CLG LAB/08-10-26/Unique Paths in
a Grid.py
7 3
28
>>>
```